

Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Blockchain for Business [IT_4424]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A hospital wants to share patient medical records securely among multiple branches. Apply blockchain principles to design a solution ensuring privacy and data integrity. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Application"
- B) A government wants to create a tamper-proof land registration system. Apply the key qualities of blockchain to design how such a system would function securely. (3)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- C) Develop a blockchain-based idea to ensure transparency in issuing land ownership certificates. (2)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 2) Compare the architectural, operational, and socio-legal effectiveness of blockchain-based land registries with traditional centralized land administration systems in maintaining data integrity, accessibility, and trust across multi-stakeholder environments. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Identify three IoT security issues that blockchain can address and explain briefly. (3)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- C) Choose an appropriate blockchain architecture to manage confidential supplier pricing data while ensuring transaction transparency across the supply chain. Justify your choice (2)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) IBM Watson integrates blockchain to validate datasets used in enterprise AI decision systems (e.g., fraud detection, risk analysis). Blockchain logs every data source and transformation step, enabling explainable and auditable AI decisions. Enterprises using Watson + Hyperledger Fabric can prove the lineage of their analytics data and models, minimizing ethical risks and data manipulation. Examine the role of blockchain in enhancing data lineage and auditability in AI analytics platforms like IBM Watson. How does this integration redefine trust and accountability in data-driven decision-making? (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) Select the most appropriate consensus mechanism to secure low-power IoT devices from (3)

cyberattacks and justify your answer.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Identify how blockchain creates both opportunities and threats in e-commerce transactions. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4) Singapore's Land Transport Authority (LTA) and IBM piloted blockchain for smart mobility payments, integrating parking, toll, and public transport billing. (5)

- A) Data from vehicles, IoT sensors, and commuters is recorded securely on blockchain. Dissect the functional role of blockchain in ensuring secure, interoperable mobility data management in Singapore's smart-city transport system.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Choose whether a private or public blockchain is more suitable for deployment on a multi-tenant cloud platform and justify your choice. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Blockchain-based games face high gas fees and energy usage during NFT minting. Solve this issue by selecting a sustainable blockchain framework. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 5) Analyze how blockchain technology enables secure and autonomous Machine-to-Machine (M2M) transactions in future IoT networks. Explain with suitable examples how machines could manage digital identities, make payments, and maintain secure records without human intervention. (5)

- A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Illustrate the difference between state channels and Plasma through a real-world application example. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) A music artist wants to prove ownership of a new song before releasing it online. How can blockchain support this artist? (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4424:Blockchain for Business

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	12
3	3	CO 3	13
4	4	CO 4	15
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4424:Blockchain for Business

#	Level	No of	Marks (%)
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		Questions	
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	11	30(60%)
4	Analysis	4	20(40%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)